

Experiment - 12

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Q1.

Design a trigger which whenever there is an insertion on student table then, the currently inserted or deleted row should be printed as it as on the output console window.

SOLUTION:

```
CREATE OR REPLACE FUNCTION fn_student_audit()
RETURNS TRIGGER
LANGUAGE plpgsql
AS
$$
BEGIN
    IF TG_OP = 'INSERT' THEN
        RAISE NOTICE 'Inserted Row -> ID: %, Name: %, Age: %, Class: %',
            NEW.id, NEW.name, NEW.age, NEW.class;
        RETURN NEW;
    ELSIF
        TG_OP = 'DELETE' THEN
        RAISE NOTICE 'Deleted Row -> ID: %, Name: %, Age: %, Class: %',
            OLD.id, OLD.name, OLD.age, OLD.class;
        RETURN OLD;
    END IF;
    RETURN NULL;
END;
$$
```

Q2.

Design a postgresql triggers that:

- Whenever a new employee is inserted in tbl_employee, a record should be added to tbl_employee_audit like: "Employee name <emp_name> has been added at <current_time>"
- Whenever an employee is deleted from tbl_employee, a record should be added to tbl_employee_audit like: "Employee name <emp_name> has been deleted at <current_time>"

The solution must use PostgreSQL triggers.

SOLUTION:

```
CREATE OR REPLACE FUNCTION audit_employee_changes()
```

```
RETURNS TRIGGER
```

```
LANGUAGE plpgsql AS
```

```
$$
```

```
BEGIN
```

```
    IF TG_OP = 'INSERT' THEN
```

```
        INSERT INTO tbl_employee_audit(message)
```

```
        VALUES ('Employee name ' || NEW.emp_name || ' has been added at ' ||
```

```
        NOW());
```

```
        RETURN NEW;
```

```
    ELSIF TG_OP = 'DELETE' THEN
```

```
        INSERT INTO tbl_employee_audit(message)
```

```
        VALUES ('Employee name ' || OLD.emp_name || ' has been deleted at ' ||
```

```
        NOW());
```

```
        RETURN OLD;
```

```
    END IF;
```

```
    RETURN NULL;
```

```
END;
```

```
$$
```

```
CREATE TRIGGER trg_employee_audit AFTER INSERT OR DELETE
```

```
ON tbl_employee
```

```
FOR EACH ROW
```

```
EXECUTE FUNCTION audit_employee_changes();
```

```
INSERT INTO tbl_employee(emp_name, emp_salary) VALUES ('Aman', 50000);
```

```
DELETE FROM tbl_employee WHERE emp_name = 'Aman';
```

```
SELECT * FROM tbl_employee_audit;
```